

- 1 (a) Newton's second law of motion states that the rate of change of momentum of a body is directly proportional to the resultant force acting on it, and in the direction of the force applied.

- (b) (i) Since resistive force is the only force acting on the car while it is braking, the resistive force is also the resultant force $F = ma = (750 \text{ kg})(4.8 \text{ m s}^{-2}) = 3600 \text{ N}$.

- (ii) Using kinematics equation,

$$v^2 = u^2 + 2as$$

$$s = 65 \text{ m}$$

- (iii) Friction exists between the tyres and the road surface.

When the brakes are applied, the wheels stop rolling. Friction between tyres and road surface causes the tyres to exert a forward frictional force on road.

According to Newton's 3rd law, the road surface will exert an equal and opposite frictional force on the tyres, causing the car to decelerate.

Note: Candidates gave directions of the forces assuming there was acceleration rather than deceleration.

- (c) (i) When the car is on a slope, the component of its gravitational force acting down the slope acts in the opposite direction to the resistive force hence reducing the resultant decelerating force. According to Newton's second law of motion, the magnitude of the deceleration thus decreases and the car travels longer before coming to rest.

Note: Lacks detail. A number stated that acceleration is greater as there was gravity down slope.

- (c) (ii) Component of car's g-force

$$mg \sin \theta = (750)(9.81) \sin 10^\circ$$

$$= 1278 \text{ N}$$

$$\text{Resultant decelerating force} = 3600 - 1278 = 2322 \text{ N}$$

$$\text{Deceleration} = 3.1 \text{ m s}^{-2}$$

Using kinematics equation

$$v^2 = u^2 + 2as$$

$$s = 101 \text{ m}$$